Kerr Landau-Siegel proof evaluation, Step 3: machine recomputation

Document evaluated: Bryce Kerr, "A simpler proof that there are no Landau–Siegel zeros," arXiv:2610.21099v1 [math.NT], 15 October 2026.

Built on: - "Step 1 - Kerr Landau-Siegel proof evaluation Step 1 18 Oct 2026 1708_v1.md" - "Step 2 - Kerr Landau-Siegel proof evaluation Step 2 18 Oct 2026 1746_v1.md"

References such as B20, FS1 or C.8 point into Step 2.

Files produced in this step (all in the folder):

File	Purpose
"Step 3 - s3_1_lattice_sums 18 Oct 2026 1752_v1.py"	Check 1: r, W, Σ i, E_p
"Step 3 - s3_2_algebra 18 Oct 2026 1752_v1.py"	Check 2: the vectors, determinant and kernel; congruences (3), (4); Lemma 4 divisibility
"Step 3 - s3_3_lemma2_basis 18 Oct 2026 1752_v1.py"	Check 3: Lemma 2
"Step 3 - s3_4_norm_bounds_and_q8 18 Oct 2026 1752_v1.py"	Checks 4 and 5: the norm constant; the q = 8 failure
"Step 3 - s3_5_margins 18 Oct 2026 1752_v1.py"	Check 6: the margin chain; "(9) implies (11)"; the order of choices
"Step 3 - script output 18 Oct 2026 1752_v1.txt"	The full output of all five scripts

The scripts need Python 3 and sympy (version 1.14.0 was used). Run each one with `python3 "<file name>"`. The script s3_5 loads s3_1 from the same folder.

What a machine check is worth under our standard: it confirms that a printed number or identity is correct, or shows that one is wrong. It is not a proof. A check over sample cases or a finite range is evidence, and is labeled as such below. The proofs that fill the breaks are the derivations in Step 2, Part C.

Results

Check 1. Lattice sums (Lemma 5; Step 2 items B20 and C.8)

Test	Method	Result
1a. Closed forms against brute force	Every (i, j, k) enumerated for D = 0 to 89, with p ∈ {11, 13, 17, 29, 97}	**All agree.** The closed forms used below are correct.
1b. r = $D^3/384 + O(D^2)$	Exact integers up to D = 4,000,000	$(r - D^3/384)/D^2$ settles at **0.0703**. Kerr's leading constant and error order are both **confirmed**.
1b. W = $5D^4/6144 + O(D^3)$	Same	$(W - 5D^4/6144)/D^3$ settles at **0.0254**. **Confirmed.**
1b. Σ i = $D^4/1536 + O(D^3)$	Same	Settles at **0.0221**. **Confirmed.** This is the main term of E_p.
1c. E_p = $D^4/(1536p) + O(D^3)$, uniformly in p	Every prime 11 ≤ p ≤ D, for D up to 20,000	The largest normalized error is **0.0010 to 0.0012**, bounded across all p. Consistent with uniformity (evidence; the proof is C.8).
1d. The ratio of leading constants	Exact fraction	(5/6144)/(1/1536) = **5/4** exactly.

Verdict: every printed asymptotic in Lemma 5 is correct.

Check 2. Algebra (Step 2 items B7, B8, B12, B29, B30)

Test	Method	Result
2a. (θ_n, σθ_n, στθ_n) = Σ $n_j\gamma_j$	Symbolic, with a = √q, b = √2 and √(2q) := ab	Difference **identically 0**. Kerr's $\gamma_1, \ldots, \gamma_4$ are correct.
2a. det(γ_1, γ_2, γ_3)	Symbolic	**4ab = 4√(2q).** Confirmed.
2a. π(v) = 0 for v = (√(2q), √2, −√q, −1)	Symbolic	**(0, 0, 0).** Confirmed. With a = √5 and b = √2 the rank is 3, so the kernel has dimension 1.
2b. (√q)^p ≡ χ(p)√q and (√2)^p ≡ (2/p)√2 (mod pO); equations (3), (4)	Exact arithmetic in O on the basis (1, √q, √2, √q√2), for 20 fundamental discriminants (−163 to 105, including −8, 40 and 88) and every odd prime p < 400 with p ∤ q. Twenty random θ_n per prime with χ(p) = −1.	**17,496 congruences checked, 0 failures.**
2c. G_{ijk}(θ_n, σθ_n, στθ_n) ≡ 0 (mod p^{⌊i/p⌋}O)	Exact arithmetic for 6 values of q, primes 11 ≤ p ≤ 29 with χ(p) = −1, D = 40	**22,500 cases, 0 failures.**

Verdict: every algebraic identity on the page is correct. Checks 2b and 2c are evidence over samples. The proofs are C.3 and

C.6.

Check 3. Lemma 2 (Step 2 items B9 and C.4)

For p ∈ {11, 13, 17, 19}, ℓ ∈ {2, 3} and D ∈ {20, 50, 100, 160}, with up to r = 12,551 monomials:

- every monomial of every G_{ijk} lies in L_D (**0 outside**);
- the change of basis is **unitriangular** when monomials are ordered by decreasing x_1 exponent.

Control (p ≤ 8, which Kerr excludes): at D = 60 the G's produce 3,009 (p = 3), 1,071 (p = 5) and 245 (p = 7) monomials outside L_D. This confirms that p > 8 is exactly what membership needs, which the paper never says (B9).

Check 4. The norm constant (Step 2 items B10, B17 and C.5)

c(|q|) = (1 + √2)(1 + √|q|)/√|q| has derivative −(1 + √2)/(2|q|^{3/2}) < 0. So its maximum over |q| ≥ 3 is at |q| = 3, where it equals **3.808** (exactly √3(1 + √2)(1 + √3)/3). Kerr's constant 8 holds with room to spare. Both ‖c_{ijk}‖ ≤ 8√|q| U in Lemma 3 and ‖θ_n‖ < B in Lemma 5 follow.

Check 5. The false step at q = 8 (FS1)

Claim on the page	At q = 8
[K : Q] = 4	√8 = 2√2, so K = Q(√2) and **[K : Q] = 2.**
1, √q, √2, √(2q) is a Z-basis of O	The elements are 1, 2√2, √2, 4. In the basis (1, √2) their coordinates have **rank 2, not 4.** Relations: √(2q) − 4 = 0 and √q − 2√2 = 0.
Lemma 3: the number of coefficients exceeds the stated bound	u = (4, 0, 0, 0) and u′ = (0, 0, 0, 1) give the **same** coefficient, 4. With 0 ≤ u_t ≤ 4 there are **273** distinct coefficients, not the 625 the proof counts.
The coordinates of v are Q-independent	v = (4, √2, −2√2, −1), and $\mathbf{v_1 + 4v_4 = 0}$.

Verdict: FS1 is confirmed. As Step 2 found, the failure is confined to q = 8 and costs nothing once |q| > 8 is assumed.

Check 6. Margins and the two inequalities that carry the argument (Step 2 items B21, B28, C.8 and C.9)

6a. Margin chain: 5/4 < 13/10 < 4/3, with 13/10 − 5/4 = **1/20** and 4/3 − 13/10 = **1/30**. Exact.

6b. "(9) implies (11)" made concrete. Since ⌊i/p⌋ > i/p − 1, hypothesis (9) gives

Σ E_p log p ≥ (13/10)(Σ i) log B − r θ(Q),

with no asymptotics needed. The script computes, exactly, the difference

(13/10 Σ i − W) log(8N√3) − r θ(Q) − r log(32r√N),

where θ(Q) is the sum of log p over primes p ≤ Q. A positive difference means (11) holds for every 3 ≤ |q| ≤ N, that is, for every γ ≥ 1: the left side of (11) only grows with B, and the right side only grows with |q|.

N	D	Q	difference / (W log B)	(11) holds
100	52	2.5	(Q < 3: Lemma 1 needs X ≥ 3)	n/a
200	134	4.8	0.0114	yes
1,000	1,154	24.2	0.0305	yes
10,000	24,893	293	0.0368	yes
1,000,000	11,556,005	60,544	0.0390	yes
10,000,000	248,966,897	958,328	0.0393	yes

All 15 grid values from N = 200 to N = 10^7 hold. The full table is in the output file. As N grows, the relative margin climbs toward the theoretical 0.04, since (1/20)/(5/4) = 1/25. This makes concrete what C.8 proves: the threshold N_0 is absolute and small.

Three limits on this check: - It covers grid points only. - The logarithms are double-precision values converted to exact fractions. The smallest margin, 1.1%, is far above rounding error. - It does not replace the proof in C.8.

6c. Order of choices in C.9. With γ = ⌈90(C_1 + 2)⌉ and ε = 1/(160C_1γ), the bracket 1/30 − (1 + C_1 + 13/20)/γ − (16/9)C_1εγ is **at least 1/90**, computed exactly for C_1 ∈ {1/10, 1, 10, 1000}. The values run from 0.0111 to 0.0130. This confirms the arithmetic of C.9. It cannot confirm the value of C_1, which no source gives explicitly.

6d. log Q. log Q − (4/3)log N tracks −2 log log N, offset by the constant log(½) − (4/3)log 3 ≈ −2.16. This confirms Kerr's log Q = (4/3)log N + O(log log N).

Summary of Step 3

1. **Every number and identity printed in the paper checks out by machine.** That covers the asymptotics of r, W and E_p, the leading constants 1/384, 5/6144 and 1/1536, the ratio 5/4, the margins 1/20 and 1/30, the vectors γ_j, det = $4\sqrt{(2q)}$, the kernel vector v, the Frobenius congruences (3) and (4), the divisibility in Lemma 4, and the triangular basis of Lemma 2.
2. **The false step FS1 is confirmed** at q = 8 on four counts, and on no other q.
3. **The one constant Kerr leaves implicit and I made explicit, 8 in Lemma 3, holds with room to spare:** the true maximum is 3.81.
4. **"(9) implies (11)" holds** at every grid point from N = 200 to N = 10^7, uniformly in $\gamma \geq 1$.
5. **Nothing here changes the Step 2 count:** 1 false step, 32 breaks (8 substantive), 3 notation defects. Machine checks confirm content. They do not supply the missing text.
6. **Still open:** B22, Lemma 6 against Masser's original, which is Step 4. And the constants C_1 and C_M, which bear on effectivity, which is Step 5.

Proposed Step 4: citation check against the originals

All three sources are now in hand.

1. **Masser 1981, Theorem 1 (p. 82)** against Kerr's Lemma 6:
    - a. Kerr allows any generators; Masser assumes a basis.
    - b. Kerr's $\mu(\Gamma)$ against Masser's $\mu(\Gamma) = \min (l - l_r)/r$. I will prove the two definitions equivalent or find where they differ.
    - c. The range "$0 \leq m_j \leq M$" against Masser's "non-negative integers not exceeding N".
    - d. Masser's hypothesis $D \geq 1$, and whether the scalar product $x \cdot y$ he uses matters.
    - e. Whether Kerr's application, with a basis because rank $\Gamma = 4$, meets every hypothesis.
    - f. Whether Lemma 6 as Kerr states it, with arbitrary generators, is true, false, or merely unsupported.

    I will read Masser's Sections 1, 2 and 5, where Theorem 1 is proved, in full.
2. **Davenport, p. 89:** whether Equation (2) and the unnumbered ζ'/ζ display support Lemma 1 exactly as used in C.1, including the primitivity hypothesis, $q \geq 3$, and $\sigma > 1$.
3. **Milne, Example 8.19:** whether the pages you supplied (274 and 275) come from the "ANTe6" file in your folder; what the example actually states; and whether it supports Kerr's congruences modulo pO, or only the Frobenius element.

The output would be "Step 4 - Kerr Landau-Siegel proof evaluation Step 4 18 Oct 2026 HHMM_v1.md", saved in the folder.